

Is Liquidity Risk Priced in Partially Segmented Markets?*

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Abstract

We analyze the impact of liquidity costs and market segmentation on asset pricing. The freely traded securities command a global market risk premium and world liquidity premia whereas the securities that can be held by only a subset of investors command additionally a conditional local market risk premium and conditional local liquidity risk premia. Our model provides a formal framework for testing important issues such as liquidity levels and risks in a realistic world market setting and for examining the interaction between investability and illiquidity. Empirical test results for a sample of thirteen emerging markets based on multiple liquidity proxies strongly support theoretical predictions.

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1 Introduction

Liquidity costs and their uncertain variation affect the willingness and the ability of foreign investors to invest in emerging markets (EMs). Unfortunately, most of the asset pricing models under barriers to portfolio flows do not take into account liquidity concerns¹. Hence, we analyze the effect of illiquidity cost and systematic liquidity risk factors on the pricing of EM securities by developing a formal international asset pricing model (IAPM) that takes into account cross-border investment barriers as well as random transaction costs. We estimate the model for major EMs to characterize the impact of liquidity on asset valuation and its interaction with investability. The model and empirical set up allow us to examine the liquidity risk premium on investable and non-investable segments of EMs.

Our new liquidity IAPM exploits the theoretical insights of international frameworks under partial segmentation. In a two country framework, we assume that all domestic securities are investable and hence can be freely traded by all investors whereas the foreign market consists of two segments: (a) a non-investable set of securities that can be held only by local investors², and (b) an investable set that can be held by all investors. We include random transaction costs on both investable and non-investable securities³. We show that the investable securities are priced as in [Acharya and Pedersen \(2005\)](#). However, the market risk and liquidity shocks are aggregated at the world level. Non-investable securities command additionally a conditional market risk premium and conditional liquidity risk premiums. The conditional market risk is a residual risk that stems from imperfect spanning of the net returns on the non-investable segment of the foreign market by net returns on substitute assets⁴. All of the sensitivity of gross returns to aggregate local market risk, sensitivity of gross returns to aggregate local market liquidity, the sensitivity of liquidity to aggregate local market risk, and the local commonality in liquidity are conditional on net returns on the substitute assets. The prices of global market risk and of global liquidity risk are the same.

¹See for example, [Stulz \(1981\)](#), [Errunza and Losq \(1985\)](#), [Eun and Janakiraman \(1986\)](#), [de Jong and Roon \(2005\)](#), [Chaieb and Errunza \(2007\)](#).

²Non-investable assets cannot be or are not traded by domestic investors due to severe explicit and/or implicit barriers. Explicit barriers include legal restrictions on ownership, foreign exchange transactions, repatriation of profits, whereas implicit barriers encompass institutional, informational, governance, and market development variables.

³Liquidity could itself be endogenously determined by other implicit barriers such as asymmetric information (see [Johnson, 2006](#); [Huang and Wang, 2009](#), among others).

⁴Substitute assets consist of all assets that can be freely traded by domestic investors including the world market portfolios, sector portfolios, closed-end country funds, exchange traded funds, ADRs and GDRs.

Similarly, the prices of local market and local liquidity risks are the same. Thus, our model extends the well-established theoretical asset pricing literature regarding the importance of liquidity for U.S. and other integrated developed markets (see, for example, [Amihud and Mendelson, 1986](#); [Pastor and Stambaugh, 2003](#); [Acharya and Pedersen, 2005](#), (AP hereafter)). AP derive a liquidity-adjusted CAPM and obtain three liquidity related betas: commonality in liquidity as in [Chordia, Roll, and Subrahmanyam \(2000\)](#), return sensitivity to aggregate liquidity as in [Pastor and Stambaugh \(2003\)](#), and liquidity sensitivity to market return.

The second key contribution of the paper is a novel empirical set up that allows the joint estimation of the diversification portfolios, DPs and of the system of pricing equations in a conditional set up. DP is the dynamic replicating portfolio obtained by projecting the non-investables returns on substitute assets returns. The weights of the DP are time-varying and computed from the conditional covariance matrices. To estimate the conditional covariance matrices we proceed in two steps. We first fit an AR(1)-NGARCH(1,1) on the assets net returns and liquidity proxy to estimate the variances. In a second step, for each investable portfolio and non-investable portfolio, we estimate a bivariate dynamic normal copula (DNC) as in [Christoffersen and Langlois \(2013\)](#). We then use the conditional variances obtained from the AR(1)-NGARCH(1,1) models and the conditional correlations obtained from the DNC model to obtain the conditional covariances needed to estimate the asset pricing model.

Our work adds to the vast empirical literature on the relation between the average stock returns and liquidity documented in several U.S. studies (see, for example, [Brennan, Chordia, and Subrahmanyam, 1998](#)), commonality in liquidity (see for example, [Chordia et al., 2000](#); [Hasbrouck and Seppi, 2001](#); [Huberman and Halka, 2001](#); [Koch, Ruenzi, and Starks, 2014](#), for evidence from the U.S.)⁵, as well as a number of empirical papers that investigate liquidity effects in international markets, (see, for example, [Domowitz, Glen, and Madhavan, 2001](#); [Lesmond, 2005](#); [Bekaert, Harvey, and Lundblad, 2007](#); [Lee, 2011](#); [Goyenko and Sarkissian, 2015](#); [Karolyi, Lee, and van Dijk,](#)

⁵Additionally, there is ample evidence that liquidity risk is unconditionally priced for the U.S. market although the contribution of the liquidity risk premium varies across studies and depends on the proxy used for liquidity, see for example, [Pastor and Stambaugh \(2003\)](#), [Acharya and Pedersen \(2005\)](#), [Sadka \(2006\)](#), [Korajczyk and Sadka \(2008\)](#), [Kim and Lee \(2014\)](#). [Watanabe and Watanabe \(2008\)](#) investigate the dynamic nature of illiquidity premium and show that liquidity betas and liquidity risk premiums are larger when investors face a higher level of preference uncertainty. [Hagstrmer, Hansson, and Nilsson \(2011\)](#) test a conditional version of the AP liquidity-adjusted CAPM and find significant time-variation in the illiquidity premium but commonality risk is the least important component.

2012)⁶. The above studies underscore the importance of liquidity in the U.S. and international markets. Our focus on pricing liquidity risk for investables and non-investables within the context of a formal asset pricing model that accounts for the partially segmented nature of a number of markets including EMs, along with the flexible empirical methodology used in the estimation, sets the paper apart from the earlier literature.

To estimate the model we require a measure of illiquidity. Liquidity is notoriously difficult to estimate especially for emerging markets with insufficient intraday transaction data. We use two proxies based on low-frequency data; the high-low spread estimator of [Corwin and Schultz \(2012\)](#) and a latent liquidity measure constructed from a principal component across three liquidity measures *à la* [Korajczyk and Sadka \(2008\)](#)⁷. The latent measure combines the information contained in different liquidity measures. The three measures are the [Lesmond, Ogden, and Trzcinka \(1999\)](#) proportion of zero returns (Zeros), Amihud’s price impact, and the high-low spread of [Corwin and Schultz \(2012\)](#)⁸.

We run the asset pricing tests with constant prices of risk. We find significant price of world covariance risk. Its magnitude of four is very plausible for the world aggregate risk aversion. The cross-sectional average of the price of local conditional covariance risk is about one with significant variation across markets. The price of local risk is proportional to the differential in risk aversion between an EM average investor and the world average investor and hence again seems economically plausible.

Our results suggest that the contribution of world liquidity risk premium to world risk premium

⁶Specifically, [Domowitz et al. \(2001\)](#) document that transaction costs in emerging markets are significantly higher than in developed markets. [Lesmond \(2005\)](#) shows that liquidity costs are higher in countries with weak legal enforcement. [Bekaert et al. \(2007\)](#) estimate an econometric model for the case of partially segmented markets and find that illiquidity predicts returns in emerging markets and that the level of liquidity is more important than systematic liquidity risk premiums. Based on an econometric model in an unconditional set up, [Lee \(2011\)](#) finds that liquidity risk is priced. [Goyenko and Sarkissian \(2015\)](#) analyze the impact of changes in the liquidity of US Treasury bills on global equity returns. [Karolyi et al. \(2012\)](#) show that commonality in liquidity is greater for emerging markets than for developed markets though it trended down for many countries. They also found that commonality in liquidity increases with capital inflows, while it decreases with market openness.

⁷[Mancini, Rinaldo, and Wrampelmeyer \(2013\)](#) and [Kim and Lee \(2014\)](#) use a similar principal component measure for exchange rates and US stocks, respectively.

⁸[Lesmond et al. \(1999\)](#) show that the Zeros measure is highly correlated with the effective spread. This measure has been applied in a number of recent papers. See for example, [Bekaert et al. \(2007\)](#), [Lesmond, Schill, and Zhou \(2004\)](#), [Lang, Lins, and Maffett \(2012\)](#), [Mei, Scheinkman, and Xiong \(2009\)](#) among others. [Levine and Schmukler \(2006\)](#) use both Zeros and the Amihud measure to examine the relationship between home market liquidity and cross listed trading across 31 home countries. [Amihud \(2002\)](#)’s widely applied price impact measure has been recently used in [Karolyi et al. \(2012\)](#) to analyze the commonality patterns of liquidity, returns, and turnover across 40 countries. See also [Fong, Holden, and Trzcinka \(2011\)](#) for a recent review of the different liquidity proxies used in international studies.

and of local liquidity risk premium to local risk premium are unconditionnaly small but the liquidity risk premiums contribution is time varying and higher in crisis periods. Unconditionally, the [Pastor and Stambaugh \(2003\)](#) covariation of returns to world market illiquidity is the dominant factor, while the [Acharya and Pedersen \(2005\)](#) covariation of illiquidity to world market returns is the dominant factor in crisis periods characterized by significant reductions in liquidity.

Finally, we demonstrate the robustness of our results to various sensitivity exercises. Using an alternative liquidity proxy, estimating the prices of risk from net returns, running the panel regression with expected liquidity costs, leaving the impact of expected liquidity costs unrestricted as in [Acharya and Pedersen \(2005\)](#), does not change our base results in any meaningful way.

The article is organized as follows. Section 2 models the liquidity risk effect on expected returns in partially segmented markets. Section 3 outlines the empirical methodology. Section 4 presents data and summary statistics. Section 5 reports the results, examines the statistical and economic significance of the liquidity risk premiums and presents a robustness check using alternative liquidity proxies and specifications of the asset pricing equations. Section 6 concludes.

2 The Asset Pricing Model

We postulate a two country global capital market: the domestic country D and the foreign country F . We label all securities that can be freely traded by all investors as investable and those that can be held by only local investors as non-investable. We can view the domestic market as a well-developed market (such as the United States) that is open to all investors and the foreign market as an EM where foreign participation is limited due to explicit and/or implicit barriers. In this setting, domestic (U.S.) investors can hold domestic securities and the investable segment of the foreign market whereas foreign (EM) investors can freely trade in the domestic and the foreign market. This characterization of the global market is fairly realistic. Of course, a market structure, in which both countries face some segmentation, is more attractive. However, our simplification makes understanding the forces at work easier without much loss of generality. Indeed, as shown by [Chaieb and Errunza \(2007\)](#), the results carry through to the more general market structure with the addition of another premium at equilibrium.

2.1 Assumptions and Notations

- **A1: Market Structure.** There are N risky assets partitioned as follow: The first N_I securities are the investable securities indexed by $i = 1, \dots, N_I$, and the second N_N assets are the non-investable securities indexed by $n = 1, \dots, N_N$. The corresponding return indices are denoted by R_I and R_N respectively. All asset returns are jointly log-normally distributed. Agents can also borrow and lend at a risk-free real rate of return, which is exogenous.
- **A2: Agents.** The economy has K agents with CRRA preferences. Agents live for one period, invest at time t , and consume at $t + 1$. Hence all investors have the same one-period trading horizon and holding period i.e. the trading frequency and holding period are set exogenously. This is a strong assumption. Allowing for endogenous trading frequency will result in less trading for illiquid assets. [Constantinides \(1986\)](#) shows that transaction costs might then have second-order effects on liquidity premia.
- **A3: Transaction costs.** Transaction costs denoted by the N -by-1 vector C_t are random and include bid-ask spread as well as search costs. They are modeled as the per-share cost of buying and selling an investable or non-investable security j . Short-selling is allowed. Both long and short holders pay transaction costs. As in AP, the one-period investors investing at t pay transaction costs proportional to the current price when closing the position at time $t + 1$. The net excess log-returns of the agents can then be written as $r_{j,t} - c_{j,t}$ for the investors who are long in asset j and as $-r_{j,t} - c_{j,t}$ for investors who are short in asset j , where $c_{j,t} = -\ln(1 - C_{j,t})$ (see [Bongaerts, Jong, and Driessen, 2011](#), for a similar assumption). We also define V as the N -by- N variance-covariance matrix of the net excess log-returns on the various securities and it can be partitioned as $V \equiv \begin{pmatrix} V_{II} & V_{IN} \\ V_{IN}^\top & V_{NN} \end{pmatrix}$, where V_{II} is the variance-covariance matrix of the investable assets, V_{NN} is the variance-covariance matrix of the non-investable securities, and V_{IN} is the covariance matrix between investable and non-investable securities. The vector of expected excess net log-returns μ and the vector of aggregate market values MV are partitioned in the same way: $\mu \equiv \begin{pmatrix} \mu_I \\ \mu_N \end{pmatrix}$ and $MV \equiv \begin{pmatrix} MV_I \\ MV_N \end{pmatrix}$.
- **A4:** The returns are measured in domestic currency, the reference currency. We follow the asset pricing literature on barriers to cross-border investments and assume that the purchasing

power parity holds (see, for example, [Stulz, 1981](#)).

2.2 Asset demands

A domestic investor d can invest in the riskless bond and the N_I investable risky assets. Let the vector $\alpha_{I,t}$ denote the fraction of his wealth invested in the risky and riskless assets at time t and γ^d is his coefficient of relative risk aversion. Investor d maximizes at time t the utility of his terminal wealth W_{t+1}^d , that is,

$$\max_{\alpha_{I,t}} E_t \left[\left(\frac{W_{t+1}^d}{1 - \gamma^d} \right)^{1 - \gamma^d} \right]$$

subject to the budget constraint $W_{t+1}^d = (1 + R_{P,t+1}^d) W_t^d$ where $R_{P,t+1}^d = \alpha_{I,t}^\top R_{I,t+1}$ is the simple return on the portfolio of investor d . In log form, the budget constraint can be written as $w_{t+1}^d = r_{P,t+1}^d + w_t^d$ where $r_{P,t+1}^d = \ln(1 + R_{P,t+1}^d)$ is the log return on the portfolio. Using the [Campbell and Viceira \(2002\)](#) discrete-time approximation of log-portfolio returns, the optimization can be restated as,⁹

$$\max_{\alpha_{I,t}} \alpha_{I,t}^\top E_t \left[r_{I,t+1} - c_{I,t+1} + \frac{1}{2} \sigma_{I,t}^2 \right] - \frac{1}{2} \gamma^d \alpha_{I,t}^\top V_{II,t} \alpha_{I,t}$$

where $\sigma_{I,t}^2$ is the vector containing the diagonal elements of $V_{II,t}$ i.e. the variance of investable asset net excess log-returns. From first order conditions, the solution for the vector of optimal portfolio weights is,

$$\alpha_{I,t} = \frac{1}{\gamma^d} V_{II,t}^{-1} E_t \left[r_{I,t+1} - c_{I,t+1} + \frac{1}{2} \sigma_{I,t}^2 \right].$$

A foreign investor f can invest in the riskless bond, the domestic securities and the foreign securities. Given his investment opportunity set, he solves a similar optimization problem as the domestic investor. The vector α_t of optimal portfolio weights for the foreign unrestricted investor can be written as,

$$\alpha_t = \frac{1}{\gamma^f} V_t^{-1} E_t \left[r_{t+1} - c_{t+1} + \frac{1}{2} \sigma_t^2 \right]$$

where σ_t^2 is the vector containing the diagonal elements of V_t i.e. the variance of investable and non-investable asset net excess log-returns and γ^f is the coefficient of relative risk aversion of the foreign investor.

⁹[Campbell and Viceira \(2002\)](#) use a Taylor approximation of the non-linear function relating log asset returns to log portfolio returns.

Using market clearing conditions and assuming that equities are in net positive supply equal to their market capitalization and the bonds are in zero net supply, we obtain the equilibrium asset pricing relationships.

2.3 Equilibrium risk and return

2.3.1 Investable set

The investable securities are priced as if the market is fully integrated. The expected excess net returns are given by,

$$E_t [r_{i,t+1}^{net}] + \frac{1}{2} var_t (r_{i,t+1}^{net}) = \gamma cov_t (r_{i,t+1}^{net}, r_{W,t+1}^{net}) \quad (1)$$

where γ is the aggregate relative risk aversion coefficient and $r_{W,t+1}^{net}$ is the net excess log-return on the world market portfolio. We then express the model in terms of gross returns,

$$\begin{aligned} E_t [r_{i,t+1}] = & -\frac{1}{2} var_t (r_{i,t+1}^{net}) + E_t [c_{i,t+1}] + \gamma [cov_t (r_{i,t+1}, r_{W,t+1}) \\ & + cov_t (c_{i,t+1}, c_{W,t+1}) - cov_t (c_{i,t+1}, r_{W,t+1}) - cov_t (r_{i,t+1}, c_{W,t+1})]. \end{aligned} \quad (2)$$

Investable securities are priced as in the liquidity-adjusted CAPM of AP with the addition of one-half the variance of excess returns. However, the market risk premiums and liquidity risk premiums are aggregated at the world level.

2.3.2 Non-investable set

To derive the equilibrium asset pricing relationship for the non-investable securities, we introduce the diversification portfolio and the hedge portfolio. The projection of the market portfolio of non-investable securities (*MPNS*) return on the space of substitute asset return is defined as the return on the diversification portfolio, *DP*, which is the portfolio of substitute assets that is most highly correlated with the market portfolio of non-investable securities. *DP* can be represented by $DP = \begin{pmatrix} V_{II}^{-1} V_{IN} M V_N \\ 0_{N_N} \end{pmatrix}$, where 0_{N_N} is the N_N -by-1 vector of zeros. The residual vector is orthogonal to the plane defined by the investable securities returns and is the return on the hedge

portfolio, *HP*. Thus, the hedge portfolio consists of a long position in the *MPNS* and a short position in *DP*. The expected net excess return on the non-investable securities at equilibrium are then given by,

$$E_t [r_{n,t+1}^{net}] + \frac{1}{2} \text{var}_t (r_{n,t+1}^{net}) = \gamma \text{cov}_t (r_{n,t+1}^{net}, r_{W,t+1}^{net}) + \left(\gamma^f - \gamma \frac{MV_N}{MV} \right) \text{cov}_t (r_{n,t+1}^{net}, r_{N,t+1}^{net} | r_{I,t+1}^{net}). \quad (3)$$

We then express the covariances in terms of gross returns

$$\begin{aligned} \text{cov}_t (r_{n,t+1}^{net}, r_{W,t+1}^{net}) &= \text{cov}_t (r_{n,t+1} - c_{n,t+1}, r_{W,t+1} - c_{W,t+1}) \\ &= \text{cov}_t (r_{n,t+1}, r_{W,t+1}) + \text{cov}_t (c_{n,t+1}, c_{W,t+1}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{W,t+1}) - \text{cov}_t (r_{n,t+1}, c_{W,t+1}), \end{aligned} \quad (4)$$

$$\begin{aligned} \text{cov}_t (r_{n,t+1}^{net}, r_{N,t+1}^{net} | r_{I,t+1}^{net}) &= \text{cov}_t (r_{n,t+1}, r_{N,t+1} | r_{I,t+1}^{net}) + \text{cov}_t (c_{n,t+1}, c_{N,t+1} | r_{I,t+1}^{net}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{N,t+1} | r_{I,t+1}^{net}) - \text{cov}_t (r_{n,t+1}, c_{N,t+1} | r_{I,t+1}^{net}). \end{aligned} \quad (5)$$

Substituting (4) and (5) in (3) we obtain,

$$\begin{aligned} E_t [r_{n,t+1}] &= -\frac{1}{2} \text{var}_t (r_{n,t+1}^{net}) + E_t [c_{n,t+1}] + \gamma [\text{cov}_t (r_{n,t+1}, r_{W,t+1}) + \text{cov}_t (c_{n,t+1}, c_{W,t+1}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{W,t+1}) - \text{cov}_t (r_{n,t+1}, c_{W,t+1})] \\ &\quad + \left(\gamma^f - \gamma \frac{MV_N}{MV} \right) [\text{cov}_t (r_{n,t+1}, r_{N,t+1} | r_{I,t+1}^{net}) + \text{cov}_t (c_{n,t+1}, c_{N,t+1} | r_{I,t+1}^{net}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{N,t+1} | r_{I,t+1}^{net}) - \text{cov}_t (r_{n,t+1}, c_{N,t+1} | r_{I,t+1}^{net})]. \end{aligned} \quad (6)$$

The extra conditional liquidity risks are

1. $\text{cov}_t (c_{n,t+1}, c_{N,t+1} | r_{I,t+1}^{net})$: the conditional commonality transaction risk between a security n and the local non-investable market N ;
2. $\text{cov}_t (c_{n,t+1}, r_{N,t+1} | r_{I,t+1}^{net})$: the conditional covariance between a security's transaction cost and the local non-investable market return;
3. $\text{cov}_t (r_{n,t+1}, c_{N,t+1} | r_{I,t+1}^{net})$: the conditional covariance between a security's return and the local non-investable market transaction cost.

As in AP, (1) affects required returns positively because investors want to be compensated for holding securities with higher transaction costs when the local market transaction costs increase; (2) and (3) affect required returns negatively because investors are willing to accept a lower return on an asset with lower transaction costs in a down local market, or an asset with higher return when local market transaction costs are high.

Note that illiquidity leads to increased expected returns. The *DP* portfolio is the one held long by some investors and short by others, but short investors need to short this portfolio to reduce their local risk exposure and long investors want to hold this portfolio as best substitute for the non-investable segment. The risk factor in our model is traded and foreign investors are homogenous.¹⁰

If the market is effectively integrated, all of the conditional market risk premium and conditional liquidity risk premiums will disappear and only global market and liquidity risk are priced as in AP. Hence perfect spanning would eliminate the emerging market valuation discount due to illiquidity risk but would not eliminate the global premiums or discounts that result from co-variation of the asset return and its transaction cost with the global market return and global transaction costs. Under complete segmentation, only local market and local liquidity risks are priced. Under partial segmentation, both global and conditional local market return and liquidity risks are priced.

Note that the equilibrium asset pricing model implies that the global risk is measured vis-à-vis the world market return (and aggregate liquidity) as the total world market portfolio is the correct portfolio to measure risk sharing.

By aggregating Equation (4) over the set of non-investable securities we obtain the conditional expected gross return on the local non-investable market portfolio,

$$\begin{aligned}
E_t[r_{n,t+1}] + \frac{1}{2}var_t(r_{n,t+1}^{net}) &= E_t[c_{n,t+1}] + \gamma[cov_t(r_{n,t+1}, r_{W,t+1}) + cov_t(c_{n,t+1}, c_{W,t+1}) \\
&\quad - cov_t(c_{n,t+1}, r_{W,t+1}) - cov_t(r_{n,t+1}, c_{W,t+1})] \\
&\quad + \left(\gamma^f - \gamma \frac{MV_N}{MV}\right)[var_t(r_{n,t+1} | r_{I,t+1}^{net}) + var_t(c_{n,t+1} | r_{I,t+1}^{net}) \\
&\quad - 2cov_t(c_{n,t+1}, r_{n,t+1} | r_{I,t+1}^{net})]. \tag{7}
\end{aligned}$$

¹⁰Impact of illiquidity on expected returns is indeterminate in [Bongaerts et al. \(2011\)](#) when the short sales are allowed and investors are heterogeneous.

This is the fundamental asset pricing equation that we will use in the empirical analysis.

3 Econometric Specification

Our empirical methodology follows two main steps. In a first step, we estimate GARCH models to obtain variances and covariances needed to estimate the asset pricing model. In a second step, we use these conditional variances and covariances in a panel regression that tests the asset pricing model jointly on the world market portfolio and all 13 emerging market non-investable and investable stock indices.

3.1 Conditional Covariance Estimation

For each asset, we estimate an AR(1)-NGARCH model on net returns

$$\begin{aligned} r_{j,t+1} - c_{j,t+1} &= \phi_{j,0} + \phi_{j,1}(r_{j,t} - c_{j,t}) + \epsilon_{j,t+1}, \quad \epsilon_{j,t+1} \sim N(0, \sigma_{j,t+1}^2) \\ \sigma_{j,t+1}^2 &= \omega_j(1 - \beta_j - \alpha_j) + \beta_j\sigma_{j,t}^2 + \alpha_j(\epsilon_{j,t} - \sigma_{j,t}\theta_j)^2 \end{aligned}$$

where $j \in \{W, N_1, I_1, CF_1^1, \dots, CF_{N_{CF}}^1, \dots, N_{N_N}, I_{N_I}, CF_1^{N_N}, \dots, CF_{N_{CF}}^{N_N}\}$, and θ_j is a parameter that captures the negative correlation between return and volatility innovations. We also estimate an AR(1)-NGARCH model for the liquidity costs c_j for all j to obtain the expected liquidity costs $E_{t-1}[c_{j,t}] = \phi_{j,0} + \phi_{j,1}c_{j,t-1}$.

For each investable and non-investable stock indices, we estimate a dynamic normal copula (DNC) as in [Christoffersen and Langlois \(2013\)](#). We estimate the correlation dynamic parameter β and α and all elements of the long-run correlation matrix Ω . Since the distribution of the GARCH shocks $\epsilon_{j,t}$ and the copula is gaussian, the joint distribution obtained by combining them is also multivariate gaussian. Hence, we use the conditional variances obtained from the AR(1)-NGARCH models and the conditional correlations obtained from the DNC model to obtain the conditional covariances needed to estimate the asset pricing model.

In order not to have a misspecified empirical setup and to be consistent with the results of the past studies (see, for example, [Carrieri, Chaieb, and Errunza, 2013](#)) that suggest that although

investables are freely available to non-local investor they still command a local risk premium¹¹, we allow investables to be priced with the local risk factor in addition to the global risk factor. We then need to form the diversification portfolios for both investables and non-investables in order to estimate the conditional covariance premia. The diversification portfolios for country n are composed of the value-weighted market portfolio W , the value-weighted portfolio of ADRs, GDRs, and direct listings S_n , and all its CFs $CF^n = \{CF_1^n, \dots, CF_{N_{CF}^n}^n\}$. We form a dynamic replicating portfolio whose allocation at time t uses conditional covariances:

$$\omega_{DP_n,t} = \Sigma_{(W,S_n,CF^n),t}^{-1} \Sigma_{n-(W,S_n,CF^n),t} \quad (8)$$

where $\Sigma_{(W,S_n,CF^n),t}$ is the covariance matrix at time t of the vector

$$r_{(W,S_n,CF^n),t} = \left[r_{W,t}, r_{S_n,t}, r_{CF_1^n,t}, \dots, r_{CF_{N_{CF}^n}^n,t} \right]^\top$$

and $\Sigma_{n-(W,S_n,CF^n),t}$ is the vector of covariances between $r_{n,t}$ and the vector $r_{(W,S_n,CF^n),t}$.

Estimating conditional covariances is complicated by the fact that the substitute assets have different time series length, with some series not overlapping at all. To ensure positive definiteness of the covariance matrix of $r_{n,t}$ and $r_{(W,S_n,CF^n),t}$, we estimate the DNC model by maximizing the log-likelihood using all available assets at each time period. This allows us to estimate one correlation model on a unbalanced panel of assets. Once an asset k becomes available, we set its first conditional correlations with other available assets l to its long-run parameter $\Omega_{k,l}$. We examine the unbiasedness of the estimators via simulations. The results suggest that the estimator of the DNC with missing values in the panel is unbiased.

The conditional variance is then computed as

$$\begin{aligned} Var_{t-1} (r_{n,t} - c_{n,t} \mid r_{(W,S_n,CF^n),t} - c_{(W,S_n,CF^n),t}) &= \sigma_{n,t}^2 (1 - II_{n,t}) \\ &= \sigma_{n,t}^2 \left(1 - \frac{Cov(r_{n,t} - c_{n,t}, r_{DP_n,t} - c_{DP_n,t})^2}{\sigma_{n,t}^2 \sigma_{DP_n,t}^2} \right) \end{aligned}$$

where $II_{n,t}$ denotes the integration index of country n , and $r_{DP_n,t} = \omega_{DP_n,t}^\top r_{(W,S_n,CF^n),t}$ is the

¹¹Carrieri, Errunza, and Sarkissian (2004) show that local factors are priced not only for emerging markets but also for developed markets.

return on the diversification portfolio.

For all assets, we estimate three different specifications of our asset pricing model. First, we consider the model in gross returns where the liquidity costs are set to zero $C_{j,t} = 0, \forall t, j \in \{W, I, N, CF\}$. Second, we use the CS spread measure of [Corwin and Schultz \(2012\)](#) estimated from the high low spread. The CS spread measures the cost of a round-trip transaction of small size, but does not capture other dimensions of liquidity such as market depth and resilience. Therefore, we follow [Kim and Lee \(2014\)](#) and use the first principal component obtained from multiple measures of liquidity. Given the availability of data in emerging markets, we use the CS spread, the proportion of zero returns, and the price impact as measures of illiquidity.

We compute the first principal component, $PCA_{j,t}$, from the standardized logarithmic changes of our three measures of liquidity and build an alternative measure of transaction cost:

$$C_{j,t}^{PCA} = \bar{C}_j^{Spread} \exp \left(\mu_j^{Spread} + \sigma_j^{Spread} \sum_{k=1}^t PCA_{j,k} \right) \quad (9)$$

where $\bar{C}_j^{Spread}$, μ_j^{Spread} and σ_j^{Spread} are respectively the average spread, the average and standard deviation of log-changes in the CS spread measure of asset j , C_j^{Spread} . Therefore, our PCA-based measure of transaction cost shares the average and volatility of the CS spread measure, but exhibits variations that reflect multiple dimensions of liquidity.

3.2 Asset Pricing Tests

To test our asset pricing model, we run panel regressions that include the world market portfolio, and the non-investable and investable stock market portfolios of all emerging markets. We use three different specifications to test our model. In the first specification, we run the panel regression with expected liquidity costs on the right hand side

$$\begin{aligned} r_{j,t} + \frac{1}{2} \sigma_{j,t}^2 &= E_{t-1} [c_{j,t}] + \alpha_j + \tau_t + \gamma Cov_{t-1} (r_{j,t} - c_{j,t}, r_{W,t} - c_{W,t}) \\ &\quad + \pi^j Var_{t-1} (r_{j,t} - c_{j,t} \mid r_{S_j,t} - c_{S_j,t}). \end{aligned} \quad (10)$$

In this regression, we estimate the country fixed effects α_j , the time fixed effects τ_t , the price of world covariance risk γ , and the prices of conditional covariance risk π^j . Note that securi-

ties S_j denote substitute assets for country j , and that by definition these conditional variances $Var_{t-1}(r_{j,t} - c_{j,t} \mid r_{S_j,t} - c_{S_j,t})$ are equal to 0 for $j = W$.

Alternatively, we test our model directly on net returns.

$$r_{j,t} - c_{j,t} + \frac{1}{2}\sigma_{j,t}^2 = \alpha_j + \tau_t + \gamma Cov(r_{j,t} - c_{j,t}, r_{W,t} - c_{W,t}) + \pi^j Var_{t-1}(r_{j,t} - c_{j,t} \mid r_{S_j,t} - c_{S_j,t}). \quad (11)$$

Admittedly excess returns net of bid-ask spreads overestimate the true cost of trading, but we use this specification as robustness check. Indeed this specification does not rely on the quality of our expected liquidity cost forecast since we model directly the expected net returns, not gross returns as in Equation (10).

Finally, we test a third specification in which we leave unrestricted the impact of expected liquidity costs as in [Acharya and Pedersen \(2005\)](#)

$$\begin{aligned} r_{j,t} + \frac{1}{2}\sigma_{j,t}^2 &= \kappa E_{t-1}[c_{j,t}] + \alpha_j + \tau_t + \gamma Cov(r_{j,t} - c_{j,t}, r_{W,t} - c_{W,t}) \\ &\quad + \pi^j Var_{t-1}(r_{j,t} - c_{j,t} \mid r_{S_j,t} - c_{S_j,t}) \end{aligned} \quad (12)$$

where κ is an additional parameter to estimate. The asset pricing model assumes that all investors have a monthly horizon, an assumption unlikely to be verified in reality. This specification captures the fact that the effect of expected liquidity costs in the data may differ from our stylized model.

4 Data

We include all emerging markets that were at some point in the MSCI Emerging Market index (Argentina, Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Israel, Jordan, Malaysia, Mexico, Morocco, Pakistan, Peru, Philippines, Poland, Portugal, Russian Federation, South Africa, South Korea, Taiwan, Thailand, Turkey, and Venezuela). We consider all equities which are traded on the major stock exchange in each country. We define a major stock exchange as the one with the highest number of equities listed. Exceptions to this rule are Brazil (Rio de Janeiro and Bovespa), China (Shanghai and Shenzhen), and Russia (Moscow and MICEX). We obtain data from Datastream. All returns are monthly, dividend-inclusive, denominated in USD, and in excess of the one-month Treasury bill rate $R_{f,t}$ obtained from Kenneth

French’s website. Depending on the country, the sample period is from March 1986 or later to December 2013.

We include data for all stocks for all countries in the MSCI All Country Index (both developed and emerging markets) to build a bottom up world market portfolio. We also use a list of Exchange Traded Funds (ETF) and country funds (CF) for each country mainly compiled from Morningstar, Bloomberg, iShares and CRSP. Returns data on ETFs and CFs are from CRSP and Datastream. We refer to both country funds and exchange traded funds as CFs and denote $CF^i = \{CF_1^i, \dots, CF_{N_i^{CF}}^i\}$ the set of CFs for country i . We match cross-listed stocks using ISINs or SEDOLs, and DRs using underlying ISINs or underlying SEDOLs obtained from Bloomberg¹².

We compute the spread measure of [Corwin and Schultz \(2012\)](#). Following their suggestion, we set negative daily spreads to zero. The monthly spread measure is the average of daily measures within the month for months in which there are at least five daily spreads available. We compute the proportion of zero returns by computing the proportion of days within a month with a zero daily total return. We set to missing the zero-return proportion for months in which less than five daily total returns are available. Following [Lee \(2011\)](#) and [Lesmond \(2005\)](#), we drop days for which more than 90% of stocks in a given country have zero returns on that day. We compute the price impact as the monthly average of daily absolute total return divided by the same days’ dollar volume. We drop observations for days with less than 100 shares traded. The monthly measure is the average daily measure for months in which there are at least five daily observations.

The size of our stock universe varies from month to month. We apply different adjustments to Datastream data. First, we set monthly total returns to missing if either the current or the previous month’s closing price in US dollars is smaller than 1. Following [Lee \(2011\)](#), we set to missing the daily returns for days t and $t - 1$ if the two-day cumulative return is smaller than 50% and at least one daily return is greater than 200%. To be included in our final sample in a given month, a stock should have a valid monthly total return, a market value at the end of the previous month, a measure of spread less than 100%, zero-return proportion, and price impact less than 5,000. Each month, we build value-weighted return, spread, zero-return proportion, and price impact indices. We construct one set of indices that contains all stock for a given country, one that contains only investable stocks in each month, and one that contains only non-investable stocks in each month.

¹²We are able to match 2,950 of the 3,729 DRs using Bloomberg

We follow [Karolyi and Wu \(2012\)](#) and characterize a stock as investable if there exists a related cross-listed stock, an ADR, or a GDR with a valid monthly return for that month.

To be able to estimate the dynamics of variances and covariances, we only keep countries for which we have at least a 10-year time series for both the non-investable and the investable stock indices. Our final sample is comprised of the following 13 countries: Argentina, Brazil, Chile, China, Israel, Malaysia, Mexico, Morocco, Philippines, Portugal, South Africa, Taiwan, and Turkey. We are currently working on adding the following set of countries to our sample: Hungary, India, Indonesia, Peru, Poland, Russia, South Korea, and Thailand.

4.1 Summary statistics

Table 1 reports summary statistics of the returns and of the monthly estimates for various liquidity measures. The emerging markets display high average returns but also high volatility as well-documented in the literature. Annualized equity volatilities range between 18% and 60%. On average, Turkey is the most volatile and Morocco the least volatile market. Many of the emerging stocks trade infrequently. The fraction of Zeros is on average twice to three times that of the world market. Stale prices create statistical biases in measured returns and measures of risk. The low volatility of the Moroccan market is likely due to a large fraction of inactive stocks.

[Insert Table 1]

We observe striking cross-country differences in the value-weighted average illiquidity of the emerging market indexes. Turkey is on average the least liquid according to the CS spread measure, while Philippines is on average the least liquid according to the Zeros and price impact measures. The smallest incidence of zero daily returns in our sample is about 10%, on average, and is observed for China and Israel. In some markets like China, the increase in the number of listed securities is positively related with the increase in market liquidity as indicated by a lower CS spread, price impact and Zeros. Notice that the number of stocks varies significantly across countries as a result of data coverage and the set of filters detailed in the Data Section. The number of stocks is quite small at the beginning of the period but increases steadily.

The PCA measure reveals that one single factor can explain on average across EMs about 20% of variation in all liquidity measures. According to the PCA measure, Philippines is the

most liquid while China is the least liquid. Comparing liquidity estimates across investables and non-investables, Panels B and C illustrate that, based on all of the four liquidity measures, non-investables are more illiquid. The difference in the average CS spread between non-investables and investables reaches 1% for Turkey. The difference in the average Zeros between non-investables and investables reaches 15% for South Africa. The average price impact is substantially higher for non-investables compared to the investables of Philippines.

Table 1 also reports the first-order autocorrelation coefficients of the different liquidity measures. All liquidity proxies exhibit strong positive autocorrelation. For most countries, the autocorrelation is larger for non-investables. The averages across non-investables (investables) of the autocorrelation of CS spread, Zeros, and price impact are respectively 0.6 (0.5), 0.5 (0.5), and 0.5 (0.3). The liquidity estimates are thus persistent although less than for US stocks. Additionally, from Table 1 we see that the contemporaneous correlation between the returns and the CS spread is negative for most countries as expected. For the other liquidity measures the contemporaneous correlation with returns is unexpectedly positive for some countries.

To further examine the difference in liquidity measures and their dynamics across investables and non-investables, we take an equally-weighted sum of each of the four liquidity measures across the investables and non-investables and plot these averages in Panels A-D of Figure 1. The plots show interesting differences across investables and non-investables. For most markets, investables are on average more liquid and less volatile. Average CS spreads are larger for non-investables compared to investables. There is no clear downward trend across the different liquidity measures for both investables and non-investables.

[Insert Figure 1]

The subprime crisis substantially affected the market liquidity of emerging markets. However the intensity differs across investables and non-investables. The dry up of liquidity is more notable for non-investables specifically with the price based and cost-volume measures. In unreported plots, we examine the time variation in the liquidity measures by country and for the investables and non-investables segments. In general, we do not see a strong downward trend in the different liquidity measures as documented for US stocks (see, for example, [Chordia, Roll, and Subrahmanyam, 2001](#), [2011](#)). However, some breaks are worth mentioning. For example, all of the liquidity measures show

a break for Brazil in 2005. Since 2005, trading in Brazil’s Bovespa has been conducted exclusively through the electronic market. We should nevertheless stress that there is no clear consensus in the literature about the effect of automated exchanges on market liquidity. During the 2007-2008 subprime crisis and Lehman bankruptcy, the spreads tripled for many countries. In some markets, the Amihud’s measure increased substantially because of tiny volume.

[Insert Table 2]

Table 2 shows the correlations among the different measures of liquidity averaged across different market segments; (1) all stocks, (2) Non-Investables and (3) Investables. With the exception of the CS spread and Zeros pair, all of the pairwise correlations are positive and range between 0.03 and 0.41. This moderate level of correlation is an indication that the three measures are not substitutes and capture different aspects of liquidity (market depth, breadth, and resilience). All of the CS spread, Zeros and price impact are positively correlated with the bid-ask spread. The correlations are overall quite similar across investables and non-investables. The measure that shows the lowest pairwise correlation is the Zeros. A major limit of Zeros is that it does not measure non-trading but rather zero returns (see Bekaert et al. (2007) for an extensive discussion of this measure). Unfortunately data limitation on the number of trades precludes us from investigating the impact of non-trading. Non-trading could also bias the CS spread. We conduct a Monte Carlo simulation to examine the effect of inactive trading on the CS spread. We find that with low number of trades per day (less than three) the probability of wrongly identifying the bid and/or ask is very high and hence the CS measure becomes biased (simulation results available on request).

5 Results

In this section, we test our asset pricing model and study whether world market and conditional market risks are unconditionally priced. We also examine the economic contribution and the dynamics of the different risk premia, specifically the global and local liquidity premia. We do this by running the panel regression on the EM investables and non-investables using the pre-estimated conditional variances and covariances as detailed in Section 3. Standard errors are clustered by time.

5.1 Prices of risk

First, we test a restricted version of our model where the price of conditional market risk is set to zero so that only global risk is priced. We test this restricted model using the CS spread measure as well as the PCA measure and the three model specifications (1) gross returns and $\kappa=1$, (2) net returns, (3) gross returns and unconstrained κ .

[Insert Table 3]

All the three model specifications show that the price of world market risk is positive and significant at the 5% level. The price of world market risk is proportional to the aggregate risk aversion. Hence, a value of three is economically meaningful. These findings are confirmed when we replace the CS spread by the PCA measure. The price of world market risk is positive, statistically significant, and has an average value of two over the three model specifications.

Using the PCA measure, the price of local risk is positive and significant in five EMs but is negative and significant for Argentina and Mexico. The average price of local risk is about 0.8 across the three model specifications. To summarize, our results are robust to the choice of the liquidity proxy but the sign and magnitude of the price of local risk changes in a few cases. It is encouraging that we find significant unconditional price of global and local risk. Nonetheless we will extend in future work the estimation by allowing for time varying prices of risk.

5.2 Liquidity risk premium

Next we examine the economic contribution of the world and local liquidity risk premia. As shown in Equation (7), the world liquidity risk premium has three components: the sensitivity of the non-investable returns to the world market-wide illiquidity; the sensitivity of the non-investable illiquidity to the world market return; and the global commonality in illiquidity. The local liquidity risk premium has two components; the conditional variance of non-investables illiquidity and the conditional covariance between non-investables index return and its illiquidity. For each country, we estimate the monthly contribution of world liquidity risk premium to the world risk premium by computing for each month, the ratio of each world liquidity component to the covariance between non-investable net returns and world market net returns. For each country, we also estimate the monthly contribution of local liquidity risk premium to the local risk premium by computing for

each month, the ratio of each local liquidity component to the conditional variance of non-investable net returns. We then compute an equally weighted average across all EMs at each time period. Figure 2 plots the time-varying contributions averaged across countries of global liquidity risk premium and of local liquidity risk premium.

[Insert Fig 2]

The averages across countries and time of the contributions of the three global liquidity risk premia amount to 0.67%, 0.42%, and 0.01%, respectively. On average, the sensitivity of non-investable returns to aggregate world illiquidity is the dominant premium, while the global commonality is the least significant among the three. The contribution of the world liquidity risk premiums varies over time. Specifically, the contribution of the sensitivity of non-investables illiquidity to world market returns is quite significant during the Asian crisis and the LTCM crisis. It is interesting to note that, unconditionally, the Pastor and Stambaugh (2003) covariation of non-investable returns to world market illiquidity is the dominant factor, while the Acharya and Pedersen (2005) covariation of non-investables illiquidity to world market returns is the dominant factor in crisis periods characterized by significant dry up of liquidity.

The averages across countries and time of the contribution of the two local liquidity risk premia amount to 0.16% and 0.73%, respectively. The conditional covariance between non-investable returns and their illiquidity is the dominant local liquidity premium. The contribution of local liquidity premium is higher in crisis periods but is still largely dominated by the local market risk premium. The average contribution of the local liquidity risk premium to the local risk premium of 1% is of similar magnitude to the contribution of the world liquidity risk premium to world risk premium. To summarize, consistent with past findings, both the world and local liquidity risk premia are quite small on average. However, note that these premia are based on value-weighted country indices. We will extend the estimation for large and small firm samples in future work to evaluate the effect of firm size on liquidity.

6 Conclusion

We develop a formal international asset pricing model that takes into account cross-border investment barriers as well as random transaction costs to analyze the effects of illiquidity cost and

systematic liquidity risk factors on the pricing of EM securities. In our model, the freely traded securities command a global market risk premium and world liquidity premia whereas the securities that can be held by only a subset of investors command additionally a conditional local market risk premium and conditional local liquidity risk premia.

We estimate the model for 13 EMs. Empirical test results support theoretical predictions. Specifically, the price of world market risk is positive, statistically and economically significant. The cross-country average price of local risk is about one, however, the sign and magnitude of the price of local risk varies across countries. Although, on average, both the world and local liquidity risk premia are quite small their importance increases during crisis periods corroborating past evidence. Our results are not sensitive to the choice of alternative liquidity proxies, estimating the prices of risk from net returns, running the panel regression with expected liquidity costs or leaving the impact of expected liquidity costs unrestricted.

In conclusion, our model provides a formal framework for testing important issues such as liquidity levels and risks in a realistic world market setting and for examining the interaction between investability and illiquidity.

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7 Figures and Tables

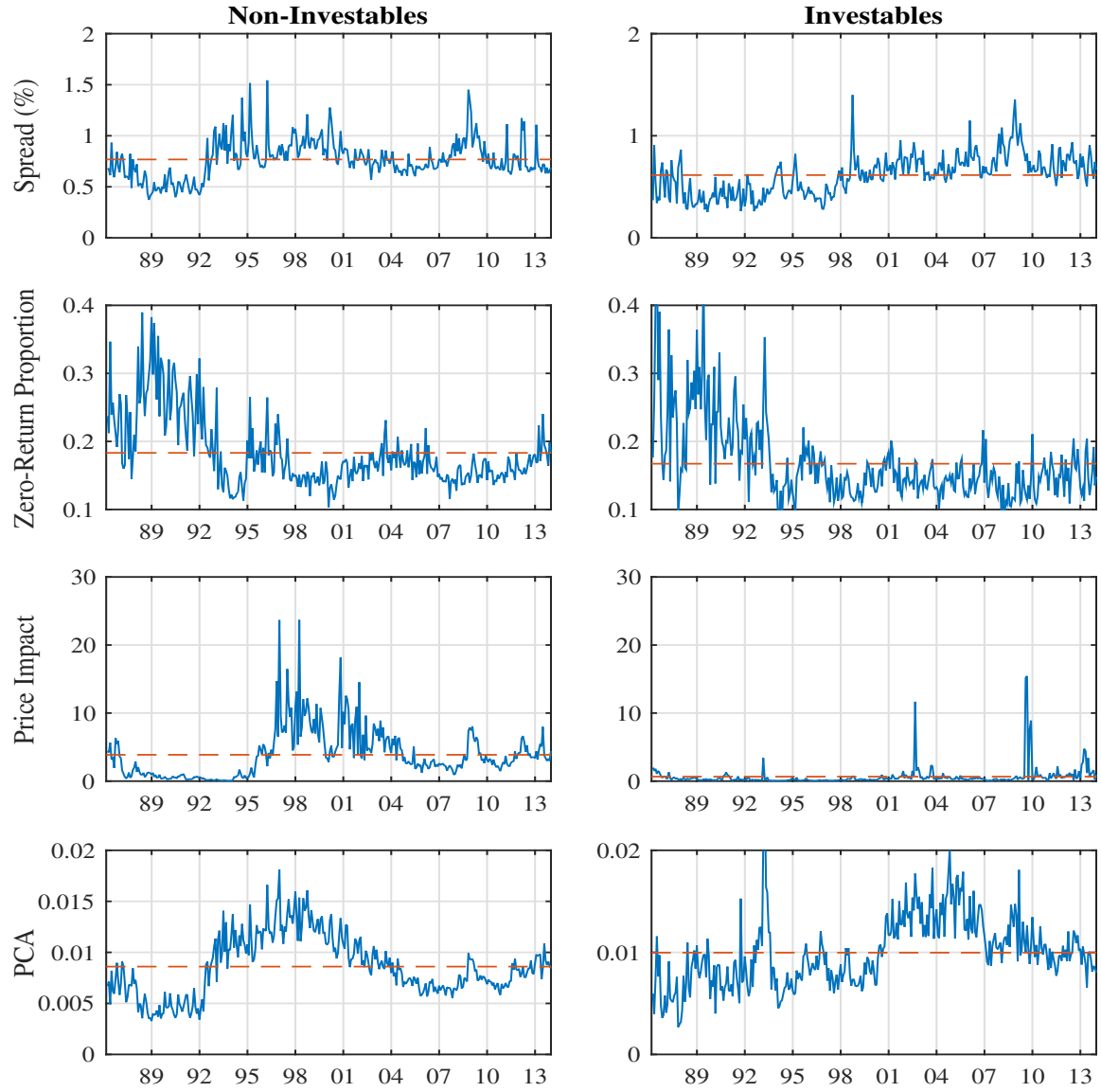


Figure 1 Liquidity measures for investable and non-investable securities averaged across emerging markets

We report monthly liquidity measures averaged across all 13 emerging markets. The first row reports on the high-low based spread measure of [Corwin and Schultz \(2012\)](#), the second on the proportion of zero-returns, the third on the price impact, and the fourth on our PCA-based measure. The left column averages the liquidity measures for value-weighted indices of non-investable securities. The right column shows the average liquidity for value-weighted indices of investable securities. We compute equally weighted averages across all available emerging markets. In each graph, the dashed line reports the time-series average of the conditional liquidity measure.

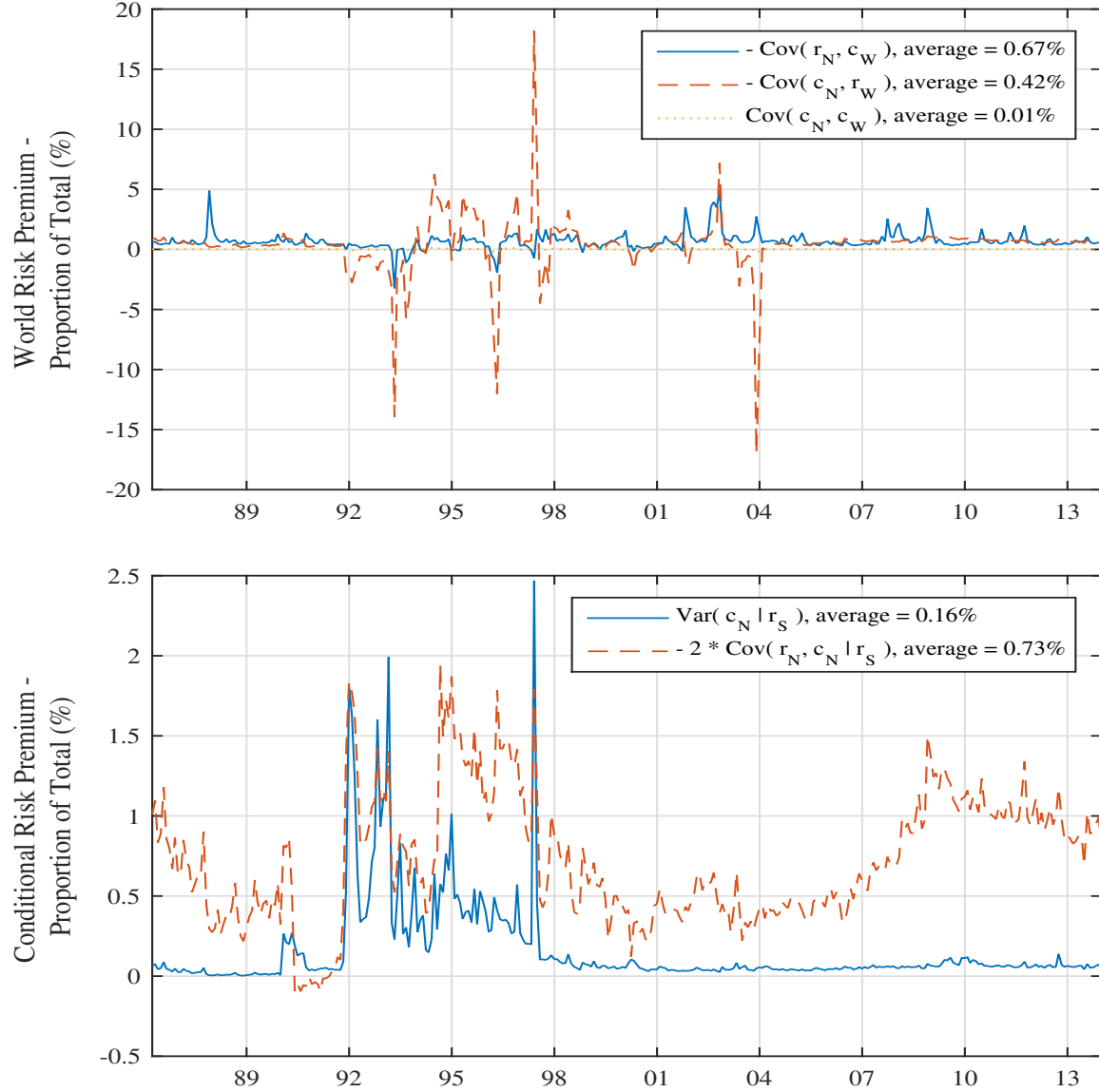


Figure 2 Contribution of liquidity covariances to the corresponding total risk premium

We report the contributions of covariance terms involving liquidity costs to the corresponding total risk premium averaged across all 13 emerging markets. The top panel reports on the world risk premium. For each country, we divide the covariances of the non-investable gross return r_N with the world market liquidity cost c_W by the covariance of net returns of the non-investable index and the world market portfolio, and then average this ratio across all 13 countries. We compute the same ratios using the covariance of the non-investable liquidity costs c_N with the world market gross returns r_W and the covariance of the non-investable liquidity costs c_N with the world market liquidity costs c_W . The bottom panel reports on the conditional local risk premium. For each country, we divide the conditional variance of the non-investable liquidity costs c_N by the conditional variance of net returns of the non-investable index, and then average this ratio across all 13 countries. We compute the same ratio for the contribution of the conditional covariance of the non-investable gross returns r_N and liquidity costs c_N . The legend in each graph reports the time-series averages of these contributions.

Table 1 - Summary Statistics

A- Returns				B- CS Spread				C - Bid-Ask spread			
Market	Annualized Mean (%)	Annualized Volatility (%)	ρ_1	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)
Panel A- All Stocks											
World	11.90	17.62	0.09	0.68	0.19	0.84	-0.25				
Argentina	12.41	34.22	0.07	0.60	0.21	0.56	-0.08	1.58	0.67	0.47	-0.27
Brazil	25.56	39.55	0.02	0.97	0.42	0.32	-0.25	2.71	2.86	0.30	0.02
Chile	20.91	27.34	0.23	0.41	0.16	0.33	-0.13	2.25	2.20	0.56	-0.23
China	20.36	46.54	0.11	0.94	0.30	0.60	0.08	0.46	0.62	0.93	0.02
Israel	12.96	24.96	0.07	0.57	0.23	0.71	-0.13	1.14	0.98	0.70	-0.27
Malaysia	14.21	30.54	0.04	0.66	0.23	0.76	-0.06	1.07	0.40	0.84	-0.16
Mexico	23.32	33.25	0.16	0.65	0.29	0.79	-0.08	1.83	1.07	0.37	-0.18
Morocco	12.86	18.30	0.06	0.34	0.20	0.83	-0.09	1.16	0.36	0.52	0.00
Philippines	21.83	35.82	0.21	0.74	0.20	0.26	-0.15	1.36	0.41	0.45	-0.22
Portugal	5.74	27.08	0.12	0.51	0.28	0.82	-0.14	0.75	0.43	0.62	-0.06
South Africa	10.70	25.18	0.07	0.63	0.28	0.80	-0.12	1.19	0.41	0.75	-0.20
Taiwan	6.21	28.29	0.05	0.71	0.17	0.67	-0.02	1.30	2.45	0.92	-0.09
Turkey	33.44	60.49	0.07	3.12	3.00	0.74	0.04	0.85	0.73	0.97	0.31
Panel B- Non-Investables											
Argentina	12.63	37.10	0.04	0.67	0.20	0.51	-0.21	1.84	0.62	0.56	-0.28
Brazil	26.18	39.67	0.02	0.99	0.42	0.32	-0.24	2.96	2.99	0.30	0.06
Chile	22.91	28.10	0.25	0.36	0.14	0.36	-0.03	2.24	1.67	0.42	-0.24
China	20.41	46.60	0.11	0.94	0.30	0.60	0.08	0.45	0.61	0.93	0.03
Israel	15.10	26.86	0.16	0.65	0.27	0.84	-0.12	1.69	1.01	0.75	-0.28
Malaysia	14.56	31.18	0.04	0.67	0.23	0.75	-0.07	1.13	0.42	0.82	-0.16
Mexico	25.12	34.82	0.16	0.64	0.29	0.73	-0.05	2.51	1.81	0.38	-0.07
Morocco	13.77	19.64	0.08	0.39	0.26	0.80	-0.08	1.59	0.49	0.68	-0.04
Philippines	21.73	38.14	0.20	0.79	0.22	0.30	-0.17	1.58	0.50	0.29	-0.20
Portugal	5.35	28.72	0.16	0.69	0.47	0.85	-0.12	1.58	0.74	0.63	-0.14
South Africa	13.84	25.26	0.09	0.57	0.24	0.91	-0.14	1.87	0.50	0.74	-0.16
Taiwan	6.51	28.29	0.04	0.73	0.17	0.68	0.00	1.33	2.44	0.92	-0.09

Turkey	32.62	60.12	0.08	3.14	2.97	0.73	0.04	0.95	0.74	0.97	0.29
A- Returns				B- CS Spread				C - Bid-Ask spread			
Market	Annualized Mean (%)	Annualized Volatility (%)	ρ_l	Mean (%)	Volatility (%)	ρ_l	Corr(r, c)	Mean (%)	Volatility (%)	ρ_l	Corr(r, c)
Panel C- Investables											
Argentina	12.51	35.76	0.06	0.57	0.25	0.55	-0.04	1.46	0.78	0.46	-0.22
Brazil	27.70	52.18	0.08	0.81	0.28	0.26	-0.13	1.25	1.34	0.06	-0.04
Chile	13.63	25.40	0.07	0.49	0.26	0.31	-0.27	2.21	2.99	0.54	-0.25
China	13.40	41.63	0.06	1.25	0.59	0.51	0.06	1.10	1.36	0.50	0.10
Israel	11.10	23.36	0.02	0.51	0.21	0.35	-0.16	0.48	0.84	0.40	-0.19
Malaysia	13.04	29.81	0.00	0.60	0.23	0.68	0.00	0.81	0.30	0.86	-0.15
Mexico	19.18	40.38	0.08	0.61	0.27	0.68	-0.17	1.70	1.11	0.37	-0.19
Morocco	8.36	18.96	0.01	0.36	0.16	0.51	-0.06	0.77	0.24	0.35	-0.05
Philippines	15.89	29.35	0.10	0.59	0.28	0.22	0.07	0.96	0.32	0.73	-0.18
Portugal	5.70	23.37	0.12	0.54	0.21	0.61	-0.28	0.49	0.26	0.26	-0.07
South Africa	9.59	26.91	0.07	0.67	0.33	0.65	-0.11	0.78	0.29	0.70	-0.20
Taiwan	7.22	29.66	0.07	0.67	0.19	0.59	-0.05	0.23	0.03	0.56	-0.14
Turkey	22.71	54.65	-0.05	2.17	2.83	0.57	0.01	0.73	0.30	0.97	0.06

Table 1 - Cont'

<i>D - Zeros</i>					<i>E - Price Impact</i>				<i>F- PCA</i>				
Market	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)	V_1
Panel A- All Stocks													
World	10.26	2.22	0.64	0.12	3.42	7.46	0.64	-0.06	0.92	0.28	0.80	-0.22	0.24
Argentina	19.59	7.28	0.51	0.01	91.19	95.47	0.69	0.11	0.65	0.25	0.58	0.09	0.24
Brazil	16.54	8.22	0.39	-0.01	128.62	215.29	0.44	-0.05	0.51	0.26	0.16	-0.17	0.22
Chile	24.65	8.13	0.37	-0.06	54.23	80.18	0.61	0.13	0.42	0.33	0.37	0.02	0.20
China	10.36	5.50	0.25	0.05	6.97	24.34	0.21	0.08	2.45E+17	3.92E+18	0.00	-0.01	0.10
Israel	10.28	2.81	0.52	0.09	169.84	256.01	0.22	0.08	3.19	1.75	0.76	-0.04	0.17
Malaysia	23.04	5.26	0.69	0.05	81.39	89.37	0.84	-0.06	0.63	0.23	0.74	-0.07	0.20
Mexico	18.32	8.42	0.72	0.08	57.90	55.99	0.23	0.00	0.91	0.32	0.59	0.10	0.19
Morocco	27.12	14.70	0.84	0.06	14.11	11.07	0.51	-0.02	2.01E+07	2.86E+08	0.00	-0.04	0.17
Philippines	29.02	9.94	0.52	-0.05	417.02	773.57	0.43	0.06	0.22	0.16	0.83	-0.05	0.24
Portugal	20.20	12.88	0.85	0.06	46.91	69.10	0.29	0.10	0.28	1.40	0.23	0.32	0.29
South Africa	18.71	8.79	0.93	-0.02	101.11	76.19	0.57	-0.04	0.87	0.44	0.83	-0.01	0.25
Taiwan	12.37	3.01	0.60	0.00	3.69	6.50	0.73	-0.08	1.34	0.65	0.90	-0.05	0.22
Turkey	17.07	5.89	0.59	0.04	3.23	3.65	0.71	-0.16	2.27	1.01	0.57	-0.07	0.23
Panel B- Non-Investables													
Argentina	19.85	5.42	0.44	0.01	101.91	108.56	0.67	0.04	0.86	0.33	0.66	-0.11	0.23
Brazil	17.84	8.57	0.40	0.03	154.41	236.61	0.49	-0.05	0.54	0.27	0.18	-0.15	0.22
Chile	27.59	8.50	0.32	-0.10	65.25	86.75	0.54	0.12	0.40	0.21	0.47	0.05	0.22
China	10.35	5.51	0.25	0.05	6.80	24.23	0.20	0.08	2.58E+17	4.13E+18	0.00	-0.01	0.10
Israel	11.69	2.75	0.46	0.09	226.52	281.17	0.20	0.09	2.44	1.18	0.85	-0.05	0.23
Malaysia	23.47	5.25	0.68	0.03	93.23	99.64	0.84	-0.06	0.65	0.23	0.74	-0.07	0.21
Mexico	22.85	8.15	0.42	0.06	141.47	230.33	0.18	0.01	0.70	0.24	0.32	0.01	0.21
Morocco	28.27	13.83	0.83	0.05	22.38	20.47	0.56	-0.05	4.00E+07	5.43E+08	0.00	-0.04	0.17
Philippines	30.12	9.51	0.44	-0.07	513.05	847.77	0.42	0.04	0.22	0.16	0.82	-0.03	0.23
Portugal	27.37	11.54	0.71	0.12	200.27	406.31	0.54	0.00	1.04	6.70	0.39	0.09	0.30
South Africa	27.74	11.69	0.95	0.05	224.19	169.00	0.62	-0.02	0.45	0.16	0.82	0.12	0.31
Taiwan	12.71	3.03	0.58	-0.03	4.97	7.13	0.70	-0.10	1.32	0.66	0.91	-0.06	0.22

Turkey	17.82	5.59	0.54	0.03	7.02	9.42	0.80	-0.15	5.01	2.16	0.70	0.08	0.23
D - Zeros					E - Price Impact				F- PCA				
Market	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)	Mean (%)	Volatility (%)	ρ_1	Corr(r, c)	V_1
Panel C- Investables													
Argentina	18.78	10.03	0.61	-0.01	84.16	129.05	0.64	0.10	1.22	0.63	0.59	0.03	0.25
	9.31	5.75	0.07	0.08	110.19	794.41	0.02	0.10	1.03	0.61	0.27	0.06	0.17
Chile	18.59	8.70	0.33	-0.10	29.83	69.89	0.05	-0.13	0.96	0.94	0.73	0.03	0.19
China	12.62	7.67	0.16	0.08	66.23	397.52	0.27	0.07	2.01	1.18	0.75	-0.11	0.19
Israel	7.39	2.66	0.20	0.02	42.13	82.52	0.37	-0.09	0.79	0.46	0.41	-0.13	0.15
Malaysia	21.21	6.86	0.58	0.09	21.74	30.02	0.77	-0.06	0.64	0.22	0.68	0.06	0.24
Mexico	14.90	8.63	0.45	-0.04	34.68	54.11	0.19	0.05	0.38	0.24	0.58	0.14	0.26
Morocco	21.94	13.92	0.55	-0.05	3.74	5.78	0.48	-0.04	1.02	0.66	0.55	-0.09	0.26
Philippines	24.93	10.99	0.55	0.02	22.87	38.63	0.35	-0.10	1.89	1.14	0.70	-0.13	0.28
Portugal	12.28	6.00	0.59	0.04	4.99	21.63	0.16	0.10	0.98	0.52	0.48	-0.09	0.20
South Africa	12.35	5.85	0.79	-0.05	21.71	25.67	0.26	-0.06	0.78	0.41	0.70	-0.03	0.27
Taiwan	11.35	3.80	0.49	0.03	0.15	0.29	0.04	-0.09	0.68	0.18	0.55	0.15	0.27
Turkey	15.88	5.97	0.50	0.05	0.85	1.27	0.72	-0.13	2.42	1.24	0.63	-0.11	0.30

Panels A, B and C of Table 1 present basic statistics of the emerging market returns for, respectively, three sets of portfolios (1) All stocks, (2) Non-investable stocks, and (3) Investable stocks. All portfolios are constructed from individual stocks as detailed in the data section of the paper. All monthly returns are denominated in USD and in excess of the one-month US T-bill rate. The panels present the annualized averages and standard deviations over the whole sample period, as well as the first-order autocorrelation. Panel A also shows the summary statistics of the world market portfolio constructed as the value weighted sum of all developed and emerging stocks.

Each panel also displays basic statistics of five liquidity proxies B) the Corwin and Schultz (2012) high-low spread (CS Spread); C) the Bid-Ask spread; D) the Lesmond, Ogden and Trzcinka (1999) Zeros measure constructed as the monthly value-weighted average of daily proportions of zero returns; E) the Amihud (2002) price impact computed as the monthly value-weighted average of daily ratios of absolute returns to dollar volumes; F) the PCA measure constructed as the first principal component extracted from the demeaned and standardized CS spread, Zeros and price impact measures.

The panels present the mean, volatility, first-order serial correlation (ρ_1) and the contemporaneous correlation with the returns. For the PCA measure, the panels also show the proportion of variance explained by the first eigenvalue termed V_1 .

The period is from March 1986 or later to December 2013.

Table 2 - Correlations between Liquidity Measures

	CS Spread	Zeros	Price Impact
<i>Panel A: All Stocks</i>			
Zeros	-0.17		
Price Impact	0.09	0.04	
Bid/Ask	0.37	0.29	0.32
<i>Panel B: Non-Investables</i>			
Zeros	-0.13		
Price Impact	0.12	0.10	
Bid/Ask	0.40	0.26	0.41
<i>Panel C: Investables</i>			
Zeros	-0.07		
Price Impact	0.11	0.17	
Bid/Ask	0.33	0.32	0.35

Panels A, B and C of Table 2 report pairwise correlations between monthly measures of liquidity averaged across all 13 emerging markets respectively for the three portfolios (1) All stocks, (2) Non-Investables, (3) Investables.

Table 3 - Prices of risk

Panel A- Liquidity adjusted World CAPM

Transaction Cost Type	<i>None</i>	<i>CS Spread</i>			<i>PCA-Based Cost</i>		
	(1)	(2a)	(2b)	(2c)	(3a)	(3b)	(3c)
Expected Liquidity Cost		1		0.52	1		-0.02
		-		0.25	-		0.14
World Covariance	1.94	3.19	3.31	2.72	1.72	1.90	1.81
t-stats	2.17	2.14	2.14	2.16	2.17	2.16	2.17
Adjusted R ²	0.33	0.33	0.34	0.34	0.33	0.33	0.33
Number of observations	6335	6335	6335	6335	6335	6335	6335

Panel B- Liquidity adjusted International CAPM with Partial Segmentation

Transaction Cost Type	<i>None</i>	<i>CS Spread</i>			<i>PCA-Based Cost</i>		
	(1)	(2a)	(2b)	(2c)	(3a)	(3b)	(3c)
Expected Liquidity Cost		1		-0.21	1		-0.18
		-		0.39	-		0.14
World Covariance	3.65	4.01	4.03	3.68	3.59	3.85	4.00
	1.92	1.89	1.89	1.88	1.90	1.90	1.90
Argentina	4.54	1.80	1.62	2.09	-2.98	-3.06	-0.93
	3.16	2.84	2.84	2.84	3.20	3.06	3.14
Brazil	1.50	1.30	1.24	1.43	1.28	1.17	1.34
	2.05	1.94	1.95	1.94	1.96	1.93	1.96
Chile	4.76	3.57	3.86	4.81	0.25	-0.21	4.16
	3.85	3.69	3.69	3.71	3.65	3.65	3.75
China	1.17	1.12	1.12	1.15	1.13	1.13	1.18
	0.65	0.64	0.64	0.64	0.65	0.65	0.65
Israel	1.96	2.10	2.26	1.86	1.67	1.67	1.56
	3.06	2.99	2.99	2.99	3.03	3.05	3.05
Transaction Cost Type	<i>None</i>	<i>CS Spread</i>			<i>PCA-Based Cost</i>		

	(1)	(2a)	(2b)	(2c)	(3a)	(3b)	(3c)
Malaysia	0.06 1.57	-0.22 1.59	-0.24 1.57	0.17 1.55	-0.26 1.51	-0.34 1.50	-0.27 1.50
Mexico	-4.73 3.33	-5.40 3.41	-5.42 3.44	-5.05 3.39	-4.09 2.85	-4.02 2.81	-4.08 2.84
Morocco	5.09 4.54	4.58 2.46	4.52 2.52	5.08 2.51	8.55 2.95	8.87 3.02	4.22 2.72
Philippines	0.00 2.56	-0.28 2.62	-0.35 2.61	0.16 2.62	0.54 2.54	0.36 2.50	-0.32 2.46
Portugal	-0.92 1.18	-1.65 1.11	-1.76 1.12	-0.77 1.16	-1.02 1.14	-1.21 1.15	-0.82 1.14
South Africa	1.32 1.70	1.34 1.55	1.44 1.56	1.38 1.59	1.45 1.79	1.36 1.83	1.29 1.75
Taiwan	1.07 1.25	0.87 1.27	0.82 1.27	1.13 1.26	0.90 1.23	0.88 1.23	0.96 1.22
Turkey	3.58 1.11	1.24 1.01	0.94 1.02	3.88 1.36	2.59 0.90	2.58 0.90	3.12 0.90
Average Local Factor	1.49	0.80	0.77	1.33	0.77	0.71	0.88
Adjusted R ²	0.35	0.34	0.34	0.35	0.35	0.35	0.35
Number of observations	6335	6335	6335	6335	6335	6335	6335

Panels A and B of Table 3 present the unconditional prices of risk factors from (A) Liquidity adjusted World CAPM and (B) Liquidity adjusted International CAPM with partial segmentation. For each model, the estimation of the panel regressions is first conducted without transaction costs. Results are in column (1). Next estimations are conducted with two measures of transaction costs: (1) the CS spread and the PCA measure. For each measure, we use three asset pricing specifications: (a) gross returns and $\kappa=1$, (b) net returns, (c) gross returns with κ set as free parameter. The t-stats are displayed under each estimation and computed from robust standard errors clustered by time. All panel regressions include country and time fixed effects. Section III of the paper details the econometric methodology used to estimate the asset pricing models.